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ARTICLE



Team learning behaviours as predictors of innovative work behaviour – a longitudinal study

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ABSTRACT

Teams have a high potential for innovation development by carrying out innovative work behaviour (IWB), but research remains limited on how engagement in team learning behaviours (TLBs) can foster IWB among team members. Especially longitudinal studies investigating the dynamic character of both constructs are missing. To clarify the relationship between TLBs and IWB over time, we conducted a three-wave longitudinal survey among interdisciplinary work teams in German vocational education. Based on a sample of 66 work teams with 275 team members consisting of vocational educators, such as teachers, school principals and social workers, a manifest cross-lagged panel model was estimated to capture the dynamic relationship between TLBs and IWB. The hierarchical structure was taken into account and missing data were imputed. The results indicate that knowledge sharing and team reflexivity at the beginning of the teamwork predict future engagement in dimensions of IWB. In the further course of the teamwork boundary spanning becomes an important TLB for IWB. The TLBs that are relevant for IWB are influenced by task interdependence and team structure. The results provide insight into how IWB can be fostered through TLBs over time. At the beginning of a work task, knowledge sharing and team reflexivity should be fostered by encouraging discussion and enriching different perspectives to reach high engagement in IWB. In the further course of teamwork information should be gathered from outside the team, for instance by inviting experts.

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Innovations can be defined as new and beneficial products or processes for organisations or their employees. As innovations become increasingly important with changes in society, competition and customer expectations, employees' innovative work behaviour (IWB) has become a key resource for organisations (Anderson, Potočník, & Zhou, 2014). IWB encompasses any employees' work activities required for innovation development (Messmann & Mulder, 2012). These work activities are carried out during the process of innovation development that spans out from generating ideas to the successful implementation of the generated ideas. This is a complex process that cannot be accomplished by individual employees alone. It follows that IWB involves social activities based on interaction with others, such as discussing a problem with colleagues.

As successful IWB requires varied knowledge, social interaction and mutual support (cf. Puranam, 2017), teams have significant potential for extensive engagement in IWB (Kim & Chung, 2017). Team settings provide an opportunity to bring together the range of knowledge, experiences and skills required to develop innovations (Fay, Shippton, West, & Patterson, 2015). Interdisciplinary teams with gender or expertise diversity are likely to be more innovative than homogenous teams (e.g., Diaz-Garcia, Gonzalez-Moreno, & Saez-Martinez, 2013; Yong, Sauer, & Mannix, 2014).

Interdisciplinary teams are set up to accomplish work tasks in many organisations (e.g., Lehmann-Willenbrock, 2017) as independent units comprising at least two individuals from different domains, embedded in the organisational context. Team members interact socially and work together on organisationally relevant tasks that require the development of novel solutions to everyday problems (cf. Kozlowski & Bell, 2003).

In case of innovation development, which depends on team characteristics, there is evidence that team learning behaviours (TLBs) such as knowledge sharing and team reflexivity can foster IWB (Widmann, Messmann & Mulder, 2016; Bednall, Sanders, & Runhaar, 2014). In other words, teams that share their knowledge, gather task-relevant knowledge, and reflect systematically are better able to explore opportunities for innovations, generate viable ideas and realise those ideas (Walter & Van der Vegt, 2013). Respective research, however, is contradictory. Whereas some studies found positive relationships other studies found no or a negative relationship between the mentioned TLBs and IWB. A reason for that could be team characteristics and domain-specific differences that the teams face. Although work teams in vocational education are comparable with other teams, such as R&D teams (cf. Huang, 2009; Liu, Schuler, & Zhang, 2013), there are some specific characteristics in the context of education, such as teachers' long-standing individualism (Vangrieken, Dochy & Raes, 2016) or high stability of the teams regarding their composition (Widmann & Mulder, 2018). Thus, studies are needed that zoom in on domain-specific characteristics of, for example vocational education (e.g., Thurlings, Evers, & Vermeulen, 2015). Moreover, previous research on the relationship between TLBs and IWB neglected the multidimensional character of IWB and included only parts of IWB (e.g., Buljac-Samardzic & Van Woerkom, 2015; Zhang, Tsui, & Wang, 2011). As the development of innovations is a complex process, and the dimensions of IWB interrelate, it is crucial to include all dimensions of IWB.

Moreover, most of the research is cross-sectional (Widmann, Messmann & Mulder, 2016). Longitudinal investigations of the relationship between TLBs and IWB are crucial, however, given the following theoretical considerations.

- (1) First of all, examination of behaviour is not possible without consideration of time (Roe, 2008). Thus, there are different factors that must be taken into account examining behaviours, such as stability and change or temporal relationships.
- (2) IWB is defined as a dynamic and context-bound construct with social-interactive and reflective facets. Most studies to date have not considered this multi-dimensional dynamic and context-bound nature of IWB, and it seems important to take these characteristics more fully into account (Messmann & Mulder, 2012). As IWB is a process extending and evolving over time (Anderson et al., 2014), and innovation

development is characterised by behavioural change and differing degrees of engagement, it is important to study those dynamics longitudinally.

- (3) Like IWB, team learning is a dynamic and temporal set of team behaviours. These can be influenced by different variables at multiple levels (organisational level, team level or individual level), generating changes or improvements over time for the team, its members and the organisation (Decuyper, Dochy, & Van den Bossche, 2010). These behaviours emerge and evolve through team members' interactions (Van der Haar, Segers, & Jehn, 2013), but there are relatively few empirical studies of behavioural change in teams (Kozlowski, 2015; Lehmann-Willenbrock, 2017).
- (4) A systematic review shows that there is also little information about how the relationship between IWB and TLBs changes over time (Widmann, Messmann & Mulder, 2016). Many cross-sectional studies have pointed to positive associations between TLBs and IWB, but no longitudinal study has yet identified reciprocal relationships (Decuyper et al., 2010). In contrast to previous studies, however, it is crucial to examine the dynamic interplay between different TLBs and dimensions of IWB, because both constructs are dynamic phenomena that change over time. There are, to our knowledge, no longitudinal studies with more than two waves exploring non-linear relationships and how such relationships may change over time (Collins, Gibson, Quigley, & Parker, 2016). Longitudinal studies with at least three waves are essential to get an accurate abstraction of reality and to investigate changes in relationships (Collins et al., 2016). On that basis, the present study examines the relationship between TLBs and IWB over time and seeks to identify any reciprocal relationships between them. The stability of TLBs and IWB is also examined to further illuminate their relationship. Deeper insight into the dynamics of these social processes can be used to foster the development of innovations in organisations. To that end, the study poses the following overarching research question: *How do team learning behaviours predict innovative work behaviour over time?*

Innovative work behaviour

The development of innovations includes, based on the models of creativity and innovation (Amabile, 1996; West, 2002), the creation as well as the implementation of innovative ideas. Whereas some studies focus only on creativity (e.g., Rahimianzarif & Moradi, 2018; Toubia, 2006) or innovation implementation (e.g., Walter & Van der Vegt, 2013), IWB considers both (Widmann, Messmann & Mulder, 2016). It incorporates all activities that need to be carried out in the work context to develop innovations. Different conceptualisations of IWB (Hammond, Neff, Farr, Schwall, & Zhao, 2011; Janssen, 2003; Thurlings et al., 2015) agree on how to describe IWB (cf. Thurlings et al., 2015). Overall, four dimensions can be derived from research that are crucial for innovation development (Messmann & Mulder, 2012). Thus IWB is a complex and dynamic construct; its four dimensions describe different activities performed that may lead to novel and beneficial products or processes.

The first of these dimensions is *opportunity exploration*. It entails activities to scan the environment for opportunities to solve problems and improve products, services, processes or strategies. The second dimension is *idea generation*. This

includes activities that can lead to new ideas about how problems can be solved, or something novel can be developed. The third dimension is *idea promotion*. It entails activities to win and organise supporters for an idea by explaining the benefits of the idea to others, discussing the required resources with colleagues and seeking permission. The fourth dimension is *idea realisation*. It contains activities to implement an idea by developing the innovation, incorporating it into regular work processes and checking and modifying the output (De Jong & Den Hartog, 2010; Messmann & Mulder, 2012).

These dimension-related activities can occur simultaneously and repeatedly (Lubart, 2001), and there is no standard sequence of either activities or dimensions; the process is not linear but chaotic and reiterative (Marinova & Phillimore, 2003). In developing an innovation, progress depends on the successful accomplishment of these activities (cf. Anderson et al., 2014) and involves forward, backward and side-ways steps (King, 1992). For instance, progress depends on finding support for realising your idea. If no support is found for an idea, it must be adapted (idea generation) and resource requirements must be checked again (opportunity exploration). The innovation process may also stagnate on one dimension of IWB. For instance, if a problem occurs during idea realisation, innovation development can stall until the problem is solved. In that situation, team members must seek support to solve the problem (idea promotion), adapt the idea and check resources and opportunities within the organisation.

Empirical findings on how IWB dimensions change and how they relate over time are missing. Additionally, little is known about any overarching patterns of IWB – how IWB activities are performed, under what conditions IWB is stable or how the four dimensions interact over time. Based on the available empirical evidence we suggest that the dimensions of IWB interact reciprocally. Carrying out the work activities of the dimensions depends on various predictors as team type and characteristics. High engagement in TLBs may be an important predictor (Widmann, Messmann & Mulder, 2016).

Team learning behaviours

Team learning describes a range of behaviours that include asking, sharing, discussing and reflecting on important methods, processes and issues. These behaviours emerge and evolve over time in the interactions between team members (Kozlowski & Klein, 2000; Van der Haar et al., 2013). Decuyper et al. (2010) identified seven core TLBs grouped into three categories that are derived from literature. (1) Basic processes include TLBs that are essential for high-quality interaction and describe what happens when teams learn, leading to change but not necessarily to improvement. (2) Facilitating processes include TLBs that are essential for evaluating, planning, experimentation etc. They help teams to learn in the right direction and make improvements more likely. They give context and focus on team learning and thus determine the effectiveness and efficiency of team interaction. (3) Storage and retrieval support persistence of team learning by enabling the team to use stored material as starting points for future tasks. This enables teams to link activities and corresponding outcomes in the past, present and future. Empirical findings provide

support for this categorisation (Widmann & Mulder, 2018). Because different relationships were found between various types of TLBs and IWB, it is important to distinguish different types of TLBs. Despite these conceptual distinctions, TLBs are interconnected throughout the categories and do not follow a linear order; it is more realistic to assume that different types of TLBs occur in varying combinations, either simultaneously or sequentially (Decuyper et al., 2010). Based on a review analysing research on the relationship between TLBs and IWB (Widmann, Messmann & Mulder, 2016) and on a cross-sectional study that preceded this study (Widmann & Mulder, 2018), we focus on three TLBs (knowledge sharing, team reflexivity and boundary spanning) that are part of basic behaviours and facilitating behaviours and are important for generating shared knowledge (cf. Decuyper et al., 2010). These TLBs are selected because knowledge is a key factor in innovation development (Dolinska, 2015), referring to understanding (including information or skills) acquired through experience, education or learning. Creating, combining, and recombining knowledge forms part of every innovation process (Dougherty, 2017).

As a basic process, *knowledge sharing* describes all those behaviours involved in sharing information, experiences and knowledge with other team members, such as communication or exchange of materials.

Team reflexivity and *boundary spanning* are facilitating processes. Team reflexivity describes team members' interaction or discussion in rethinking strategies, methods, tasks, processes and so on to recombine their knowledge and to develop a clear vision of goals, methods and the status of the team task. Boundary spanning refers to all behaviours that relate to gathering information, knowledge and experience from outside the team – for instance, inviting experts or communicating with colleagues who are not team members.

Existing evidence supports the importance of knowledge for innovation development and confirms the positive association between these TLBs and elements of IWB (Widmann, Messmann & Mulder, 2016; Somech & Khalaili, 2014; Walter & Van der Vegt, 2013). By gathering information from outside the team and sharing knowledge within the team, these TLBs generate a broad knowledge base; teams that reflect on team processes are able to recognise when information is needed from outside the team.

Dynamics of innovative work behaviour and team learning behaviours

Communication between individuals and collective knowledge management are essential for innovation development (Dolinska, 2015). Given the social nature of innovation development, teams have significant potential for extensive engagement in IWB. However, the execution of many IWB activities depends on the extent of engagement in TLBs. To exploit the advantages of teams over individuals, such as a broader knowledge base, team members must engage in TLBs to build common extensive knowledge and shared understanding of their tasks and strategies for innovation development (Müller, Herbig, & Petrovic, 2009). Cross-sectional relationships between knowledge sharing, team reflexivity, boundary spanning and IWB are well documented (Widmann, Messmann & Mulder, 2016), and we assume that TLBs also predict future IWB.

Team members can use the extensive knowledge generated by TLBs for innovation development. Because at the beginning of team work nobody knows who has which

knowledge, a team member has no access to the individual knowledge of other team members, which must be made available for other team members through knowledge sharing (Zhang et al., 2011).

At the beginning of work task knowledge sharing is crucial for subsequent just-in-time use during innovation development, enabling team members to explore opportunities because they are better able to identify the organisation's needs and to analyse problems. This ability is also important for generating a fully developed idea (Nemeth & Owens, 1996). Team members know who has the relevant expertise and network within an organisation that can be used for idea promotion. Knowledge sharing also enhances team members' ability to evaluate alternative solutions during idea realisation (Somech & Drach-Zahavy, 2007).

In some cases, the individual's knowledge may be implicit, which means it cannot be communicated directly because the individual is not aware of it; neither it is available to other team members. However, by explicating and reflecting on implicit knowledge, it can be accessed for process regulation during innovation development (Müller et al., 2009). Team reflexivity on work experiences can enhance team members' flexibility in responding to novel challenges (Messmann & Mulder, 2015). Team members build episodic cases of experienced situations that facilitate knowledge transfer to novel and unexpected situations and can improve their IWB (Kolb, 1984; Kolodner, 1997). This flexibility and ability to transfer knowledge can broaden the scope of actions for future idea promotion and idea realisation. Highly reflective teams also have a clear vision of the status quo, what they want to achieve and how to do so (Decuyper et al., 2010). This kind of clarity at the beginning of team work supports innovation development by enabling teams to recognise innovation opportunities and to generate new ideas. Clarity about goals also enables teams to control their activities during idea realisation. In general, a high level of engagement in team reflexivity encourages constructive communication that generates inputs for the development of new ideas (Buljac-Samardzic & Van Woerkom, 2015).

Reflection also enables the team to identify sources of relevant expertise and knowledge beyond the team itself. By gathering information from outside the team, the knowledge base, generated by sharing and team reflexivity, can be extended. Communicating with people outside the team provides new information and other perspectives on the task or problem, fostering generation of new ideas and the adoption of new ways of doing things (Somech & Khalaili, 2014).

Based on these considerations, we assume that a high level of engagement in knowledge sharing, team reflexivity and boundary spanning positively predicts IWB, and we formulate the following hypotheses.

H1: Knowledge sharing positively predicts a) opportunity exploration, b) idea generation, c) idea promotion and d) idea realisation.

H2: Team reflexivity positively predicts a) opportunity exploration, b) idea generation, c) idea promotion and d) idea realisation.

H3: Boundary spanning positively predicts a) opportunity exploration, b) idea generation, c) idea promotion and d) idea realisation.

Method

Procedure and sample

Over the course of one school year, we conducted a questionnaire study in three waves (October 2015, February 2016 and June 2016). We contacted 339 vocational colleges in Bavaria, Germany, by phone, mail and presented the research in school meetings. Two hundred sixty-two colleges had no interest in participation. Eventually, 124 teams from 77 colleges agreed to participate and met our selection criteria. All teams were selected based on interviews with school principals about team characteristics and team tasks. All teams had to fit to our definition of work teams: they are embedded within vocational colleges as independent units. Their members interact socially and work together on organisationally relevant tasks (cf. Kozlowski & Bell, 2003). Furthermore, the teams had to be interdisciplinary in the sense that members have diverse qualifications, experiences and work responsibilities. Interdisciplinary teams were selected because such teams exhibit cognitive diversity that can lead to cognitive conflicts (Piaget, 1977). Thus, team learning is more likely to occur in interdisciplinary teams than in disciplinary teams. Finally, the teams were selected in accordance with our definition of innovations. All teams work on complex, knowledge intensive work tasks to address various work-related challenges that require novel solutions to improve the current situation.

Tasks varied from the integration of refugees to quality assurance, and had to be completed within one school year, making it essential to collect all data within that timeframe. During that period, the teams remained relative stable in terms of their composition and structure, which helped to reduce the number of confounding variables that might influence the relationship between TLBs and IWB. Because of dropout (during T1 = 117 work teams participated with 593 team members; at T2 = 106 work teams with 409 team members; at T3 = 87 work teams with 315 team members) the final sample comprised 275 vocational educators (52.4% male, 41.1% female), such as teachers, social workers and school principals in 66 interdisciplinary work teams (response rate at the individual level = 46.4 %). The participants' average age was 44.8 years ($SD = 9.86$). Team size ranged from 2 to 23 persons ($M = 7.75$; $SD = 3.82$).

Measures

To match the conceptualisations of the core constructs outlined above, variables were selected based on the results of a systematic literature review and a cross-sectional study investigating the relationship between TLBs and IWB (Widmann, Messmann & Mulder, 2016; Widmann & Mulder 2018). They were measured using existing validated scales which were used in the original format. A small number of items were slightly adapted to the domain of vocational education. For instance, the word *organisation* was replaced with *college*. TLBs and IWB were measured simultaneously at each of the three measurement points, using identical instruments. Controls were measured at the first measurement point. Adequate internal consistency (Cronbach's α) was achieved for all scales (see Table 1).

Dependent variable

IWB was assessed using Messmann and Mulder's (2012) 28-item scale. In accordance with our conceptualisation, this instrument takes into account the dynamic, context-bound and multidimensional character of IWB. Each dimension was assessed by seven items, adapted to the vocational education context. The items described concrete work activities, including social-interactive and reflective activities during the innovation process, such as 'Keeping up with structures and processes in the college' and 'Keeping up with the latest developments in the college' (opportunity exploration), 'Discussing one's own ideas for changes with close colleagues' and 'Exchanging ideas for concrete changes at work with close colleagues' (idea generation), 'Recruiting colleagues for actively supporting the realisation of an idea' and 'Recruiting superiors for actively supporting the realisation of an idea' (idea promotion), 'Thinking carefully about the goals that should be attained through the realisation of an idea' and 'Systematically reflecting on experiences gained during the realisation of an idea' (idea realisation). Responses were based on a six-point Likert-type scale ranging from 1 = *never* to 6 = *nearly always*.

Independent variables

Knowledge sharing was assessed on an eight-item scale consisting of five items from Staples and Webster (2008) and three additional items from Neumann (2017): (a) 'In this team, we exchange our practical experiences regarding the common task'; (b) 'In this team, we share project relevant knowhow'; (c) 'In this team, we share our previous knowledge'. Responses were based on a five-point Likert-type scale ranging from 1 = *never* to 5 = *very often*.

Team reflexivity was assessed using Schippers, Den Hartog, and Koopman's (2007) scale. As an example, three of the 23 items were (a) 'We regularly discuss whether the team is working effectively'; (b) 'We review our methods of working as a result of changes in the environment'; (c) 'We reflect on the way in which we communicate'. Responses were based on a five-point Likert-type scale ranging from 1 = *never* to 5 = *very often*.

Boundary spanning was assessed using Hirst and Mann's (2004) four-item scale, including such items as (a) 'When necessary, team members consult with individuals who possess knowledge relevant to the project'; (b) 'Team members seek relevant information from across the college'. Responses were based on a five-point Likert-type scale ranging from 1 = *never* to 5 = *very often*.

Controls

Based on previous research on team learning, we added two important control variables related to project characteristics (Bunderson & Boumgarden, 2010; Van den Bossche, Gijsselaers, Segers, & Kirschner, 2006).

Team structure was assessed using Bresman and Zellmer-Bruhn's (2013) five-item scale, which contains items such as (a) 'Team members are sure about what they are expected to do'; (b) 'The team spends time making sure every team member understands the team objectives'.

Task interdependence was assessed using a seven-item scale that included five items from Rupperecht (2014) and two additional items from Neumann (2017): (a) 'To

Table 1. Means, standard deviations, reliabilities and correlations of the variables.

	M	SD	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
1 TI	3.98	.59	.84																						
2 TS	3.87	.69	.45*	.77																					
3 KS1	4.21	.61	.37*	.57*	.80																				
4 KS2	4.28	.56	.30*	.39*	.39*	.77																			
5 KS3	4.24	.60	.25*	.35*	.23*	.56*	.77																		
6 TR1	3.72	.59	.46*	.67*	.63*	.39*	.32*	.94																	
7 TR2	3.70	.62	.28*	.45*	.35*	.72*	.44*	.50*	.95																
8 TR3	3.69	.64	.30*	.42*	.21*	.45*	.73*	.48*	.63*	.95															
9 BS1	3.81	.75	.38*	.59*	.33*	.19*	.60*	.31*	.56*	.37*	.18*	.72													
10 BS2	3.71	.87	.20*	.30*	.28*	.57*	.30*	.31*	.42*	.40*	.37*	.40*	.87												
11 BS3	4.60	.90	.31*	.37*	.34*	.40*	.50*	.46*	.42*	.63*	.28*	.32*	.32*	.84											
12 OE1	4.88	.73	.35*	.47*	.39*	.24*	.23*	.50*	.29*	.34*	.44*	.28*	.55*	.82	.51*										
13 OE2	4.89	.77	.27*	.36*	.36*	.44*	.19*	.39*	.51*	.30*	.31*	.44*	.53*	.51*	.85	.39*									
14 OE3	3.70	.83	.17*	.31*	.21*	.36*	.62*	.33*	.36*	.59*	.32*	.47*	.52*	.39*	.28*	.86	.37*								
15 IG1	4.78	.82	.34*	.35*	.30*	.24*	.20*	.37*	.30*	.25*	.32*	.18*	.45*	.73*	.33*	.26*	.88	.37*							
16 IG2	4.77	.86	.37*	.31*	.31*	.39*	.17*	.33*	.47*	.24*	.27*	.34*	.46*	.36*	.76*	.18*	.37*	.91	.41*						
17 IG3	4.88	.78	.32*	.35*	.35*	.35*	.49*	.40*	.33*	.48*	.36*	.34*	.82*	.59*	.60*	.50*	.41*	.41*	.90	.47*					
18 IP1	4.66	.92	.37*	.33*	.29*	.23*	.20*	.41*	.27*	.30*	.39*	.28*	.50*	.75*	.47*	.32*	.79*	.35*	.85*	.47*	.91				
19 IP2	4.61	.99	.36*	.29*	.31*	.34*	.14	.29*	.45*	.23*	.27*	.36*	.48*	.43*	.81*	.24*	.35*	.85*	.48*	.47*	.91	.92			
20 IP3	4.80	.82	.29*	.29*	.19*	.31*	.50*	.30*	.31*	.51*	.22*	.28*	.83*	.53*	.48*	.51*	.51*	.46*	.82*	.48*	.46*	.92	.45*	.89	
21 IR1	4.59	.85	.36*	.49*	.37*	.26*	.23*	.57*	.37*	.41*	.38*	.27*	.57*	.78*	.41*	.31*	.73*	.36*	.49*	.76*	.39*	.45*	.89		
22 IR2	4.65	.85	.30*	.34*	.27*	.34*	.22*	.36*	.60*	.37*	.26*	.42*	.50*	.39*	.79*	.22*	.29*	.81*	.42*	.42*	.83*	.40*	.46*	.90	
23 IR3	4.63	.93	.23*	.25*	.23*	.31*	.43*	.29*	.28*	.44*	.24*	.23*	.80*	.52*	.53*	.46*	.40*	.37*	.80*	.50*	.50*	.83*	.40*	.37*	.91

Note. N = 216 to 275 (due to missing values); values for Cronbach's alpha are presented in the diagonal of the correlation matrix. TI = task interdependence; TS = team structure; KS = knowledge sharing; TR = team reflexivity; BS = boundary spanning; OE = opportunity exploration; IG = idea generation; IP = idea promotion; IR = idea realisation. Numbers after the variable name refer to measurement points; *p < .01.

succeed in handling tasks in the team, it is necessary that all team members do a good job'; (b) 'To succeed in handling tasks in the team, I must collaborate closely with the people in my team'. For both variables, responses were based on a five-point Likert-type scale ranging from 1 = *does not apply at all* to 5 = *definitely applies*.

Individual-level controls included gender, age, and team stability.

Analyses

To capture the dynamic relationship between TLBs and IWB, a manifest cross-lagged panel model was estimated. It includes on the one hand cross-sectional correlations between the variables at T2 and T3, autoregressive regression paths for each variable across time and cross-lagged regression paths between T1 and T2. To account for the hierarchical structure of the data, the following procedure was implemented (similar to the routine implemented in Mplus): (1) the model was estimated with the development version of lavaan 0.6 (Rosseel, 2017) using Maximum Likelihood estimation and robust standard errors; (2) to impute missing data, the resulting object was then passed to semTools 0.4 (Jorgensen, Pornprasertmanit, Schoemann & Rosseel, 2017), which utilises mice 2.30 (Van Buuren & Groothuis-Oudshoorn, 2016) for data imputation. Based on recommendations from the methodological literature and the moderate amount of missing data, ten imputed datasets were created (Enders, 2010). (3) The sampling design (simple random sampling on team-level) was then specified, and the lavaan-model was re-estimated with the imputed data in lavaan.survey 1.1.3.1 (Oberski, 2014), using Maximum Likelihood estimation. The standard errors, chi-square statistic, and all derived fit indices and statistics are adjusted for the complex sampling design, which is equivalent to Pseudo-Maximum Likelihood (PML; Oberski, 2014). The scaling correction factor (i.e., the design effect) was 2.236, clearly justifying the analytic approach chosen.

Results

Descriptive statistics and correlations

Mean scores indicate that the teams engaged strongly over the year in all dimensions of IWB and in all TLBs. The teams judged their team structure as clear and their task interdependence as high. The means of all variables were stable over time with the exception of opportunity exploration and boundary spanning; between T2 and T3, opportunity exploration decreased while boundary spanning increased.

All TLBs relate to the four dimensions of IWB (see Table 1), with the exception of idea promotion T2 and knowledge sharing T3, where no significant correlation was found. The results indicate positive relationships both between the TLBs and between the several dimensions of IWB. With regard to control variables, both task interdependence and team structure correlated positively with TLBs and the dimensions of IWB.

Cross-lagged panel model

The model shows an adequate fit to the data, as indicated by the common fit indices ($\chi^2 = 51.02$, $p = .218$; $df = 44$; SRMR = .05; CFI = .99; TLI = .99; RMSEA = .02). In light of

the relatively small sample size and the known sensitivity of the χ^2 -test to sample size, we recalculated the χ^2 -statistic based on the small sample correction developed by Yuan (2005; as outlined in McNeish, 2017). The small sample-corrected χ^2 -test was $\chi^2 = 49.137$, $p = .26$; thus, the model still shows good fit to the data after taking into account the sample size. The autoregressive paths (see Tables 2 and 3) indicate stability of the variables. Opportunity exploration, idea promotion and idea realisation predict respective behaviour between T1 and T2 but not between T2 and T3. Idea generation at T1 does not predict idea generation at T2, but idea generation at T2 predicts idea generation at T3. Team reflexivity and knowledge sharing predict behaviour between T1 and T2 as well as between T2 and T3. Boundary spanning at T1 predicts boundary spanning at T2 but not from T2 on T3, when it increases, as shown by the descriptive statistics (see Table 1).

We also found statistically significant longitudinal predictions both between the different TLBs and between the dimensions of IWB that differ between the first half year (T1 – T2; see Table 2) and the second (T2 – T3; see Table 3). We found both negative and positive relationships. Idea generation relates negatively to the other dimensions at T2 (β ranges from $-.19$ to $-.21$). Idea generation at T2 relates negatively to idea realisation ($\beta = -.29$, $p < .05$)

Table 2. Cross-lagged regression paths of the model between T1 and T2.

	Innovative Work Behaviour				Team Learning Behaviours		
	OE2	IG2	IP2	IR2	BS2	KS2	TR2
OE1	.32** (.09)	-.03 (.13)	.19 (.13)	-.03 (.10)	.00 (.11)	-.12 ⁺ (.07)	-.13 ⁺ (.08)
IG1	-.20* (.08)	.11 (.10)	-.19 ⁺ (.11)	-.21* (.10)	-.15 (.11)	.11 ⁺ (.07)	.12 (.07)
IP1	.30** (.08)	.23* (.10)	.43** (.11)	.27* (.09)	.19* (.08)	.02 (.06)	-.02 (.07)
IR1	-.05 (.08)	.00 (.11)	.04 (.11)	.35** (.10)	.04 (.13)	-.05 (.07)	.06 (.07)
BS1	-.04 (.08)	-.02 (.08)	-.02 (.10)	-.02 (.09)	.37** (.08)	.05 (.05)	.01 (.06)
KS1	.20* (.10)	.20 (.12)	.29* (.13)	.09 (.09)	-.03 (.11)	.14* (.07)	-.02 (.08)
TR1	.09 (.09)	.10 (.13)	-.10 (.14)	.03 (.12)	.08 (.15)	.20* (.10)	.44** (.10)
TI					.05 (.10)	.10 ⁺ (.05)	.02 (.05)
TS					.06 (.12)	.09 ⁺ (.05)	.09 ⁺ (.05)

Note. Standardised β and SE (in brackets); ⁺p .10, *p < .05, **p < .01. TI = task interdependence; TS = team structure; KS = knowledge sharing; TR = team reflexivity; BS = boundary spanning; OE = opportunity exploration; IG = idea generation; IP = idea promotion; IR = idea realisation.

Table 3. Cross-lagged regression paths of the model between T2 and T3 and second-order cross-lagged regression paths between TLBs at T1 and IWB at T3.

	Innovative Work Behaviour				Team Learning Behaviours		
	OE3	IG3	IP3	IR3	BS3	KS3	TR3
OE2	.06 (.11)	.49** (.09)	.14 (.10)	.43* (.13)	.19 (.12)	-.06 (.07)	-.04 (.08)
IG2	.08 (.09)	-.18 ⁺ (.09)	.17 ⁺ (.10)	-.29* (.12)	-.04 (.11)	.03 (.07)	-.05 (.06)
IP2	.17 ⁺ (.10)	.16* (.08)	.15 (.10)	.44** (.11)	.12 (.11)	-.05 (.07)	-.01 (.06)
IR2	-.19 ⁺ (.11)	-.05 (.10)	-.10 (.10)	-.17 (.12)	.17 (.11)	.02 (.07)	.07 (.07)
BS2	.30** (.07)	.02 (.07)	.06 (.07)	-.05 (.06)	.04 (.06)	.02 (.04)	.07 ⁺ (.04)
KS2	.14 (.13)	.15 (.11)	.13 (.12)	.22 ⁺ (.12)	.19 (.12)	.55** (.09)	.03 (.08)
TR2	.06 (.13)	-.19 (.13)	-.07 (.13)	-.08 (.15)	-.08 (.14)	-.02 (.09)	.44** (.09)
BS1	.11 (.09)	.08 (.06)	.02 (.07)	.07 (.08)	-.10 (.08)	-.07 (.07)	-.14** (.05)
KS1	-.17 ⁺ (.10)	-.03 (.09)	-.13 (.11)	-.12 (.10)	.05 (.11)	-.04 (.08)	-.08 (.07)
TR1	.25* (.11)	.27* (.11)	.24 ⁺ (.13)	.21 ⁺ (.13)	.51** (.12)	.15 ⁺ (.09)	.39** (.09)
TI					-.02 (.04)	.07 ⁺ (.04)	.08 ⁺ (.04)
TS					-.06 (.04)	.05 (.06)	.03 (.05)

Note. Standardised β and SE (in brackets); ⁺p .10, *p < .05, **p < .01. TI = task interdependence; TS = team structure; KS = knowledge sharing; TR = team reflexivity; BS = boundary spanning; OE = opportunity exploration; IG = idea generation; IP = idea promotion; IR = idea realisation.

and positively predicts idea promotion ($\beta = .17, p < .10$) at T3. Idea promotion at T1 predicts opportunity exploration, idea generation and idea realisation at T2 (β ranges from .23 to .43), and idea promotion at T2 predicts the other dimensions at T3 (β ranges from .17 to .44).

With regard to the TLBs, only two significant predictions were found. Team reflexivity at T1 predicts knowledge sharing at T2 ($\beta = .20, p < .05$) and boundary spanning at T2 predicts team reflexivity at T3 ($\beta = .07, p < .05$). The second-order paths between T1 and T3 indicate that team reflexivity predicts boundary spanning ($\beta = .51, p < .01$) and knowledge sharing ($\beta = .15, p < .10$) at T3 while boundary spanning at T1 relates negatively to team reflexivity ($\beta = -.14, p < .10$) T3.

The results indicate positive significant cross-sectional correlations between all TLBs and all dimensions of IWB ($p < .01$). The correlations between TLBs range from $r = .16$ to $r = .20$ at T2 and from $r = .12$ to $r = .17$ at T3. The correlations between dimensions of IWB range from $r = .34$ to $r = .54$ at T2 and from $r = .16$ to $r = .43$ at T3. The correlations between TLBs and IWB range from $r = .08$ to $.21$ at T2 and from $.11$ to $.42$ at T3.

At T2, R^2 ranges from .18 for idea generation to .29 for team reflexivity and at T3 from .21 for idea promotion and .46 for team reflexivity.

Hypotheses testing

We hypothesised that all TLBs (knowledge sharing, team reflexivity and boundary spanning) would positively predict the four IWB dimensions (opportunity exploration, idea generation, idea promotion and idea realisation). The significance of predictions linking TLBs and IWB differ from the first half-year (T1–T2, see [Table 2](#)) to the second (T2–T3, see [Table 3](#)).

In the first half of the school year, knowledge sharing at T1 predicts opportunity exploration at T2 ($\beta = .20, p < .05$) and idea promotion at T2 ($\beta = .29, p < .05$). These results provide support for H1a und H1c. Reciprocally, a negative relationship was found between opportunity exploration T1 and knowledge sharing T2 ($\beta = -.12, p < .10$) and team reflexivity T2 ($\beta = -.13, p < .10$). Idea promotion at T1 predicts boundary spanning at T2 ($\beta = .19, p < .05$).

In the second half of the year, boundary spanning at T2 positively predicts opportunity exploration at T3 ($\beta = .30, p < .01$), and knowledge sharing at T2 predicts idea realisation at T3 ($\beta = .22, p < .10$). These results provide support for H1d und H3a. No significant relationship was found between team reflexivity and boundary spanning at T2 and the dimensions of IWB at T3 or for the effects of IWB dimensions on TLBs. Only the described effects of TLBs on dimensions of IWB were statistically significant (but not vice versa).

Regarding the prediction of behaviours at the end of school year (T3) from those at the beginning (T1) (see [Table 3](#)), relationships were found between team reflexivity at T1 and opportunity exploration ($\beta = .25, p < .05$), idea generation ($\beta = .27, p < .05$), idea promotion ($\beta = .24, p < .10$) and idea realisation ($\beta = .21, p < .10$) at T3. These results provide support for H2a–H2d. We also found a negative relationship between knowledge sharing at the beginning of school year and opportunity exploration at the end ($\beta = -.17, p < .10$). These results fail to provide support for H1a.

Discussion

The aim of the study was to examine the role of TLBs as predictors of IWB and to identify reciprocal relationships between TLBs and IWB. The study provides insights

that suggest how IWB can be fostered. Knowledge sharing and team reflexivity may provide a starting point to fostering the engagement in IWB. During team work more attention should be paid to boundary spanning. In the following section, changes in TLBs and IWB are discussed first. After that, the relationships between TLBs and the dimensions of IWB are discussed, including the changes of different relationships over time.

Changes in TLBs and in IWB

First, the results indicate high levels of engagement in TLBs and IWB. Second, the teams exhibited behavioural stability throughout the year for TLBs, except for boundary spanning. One possible reason for the high engagement and stability relates to the selection of teams, all of which worked on complex work tasks to meet challenges, and were together longer, with little change in team composition. The results might be different for recently established teams, as more team learning occurs in the later phases of team development (Raes, Kyndt, Decuyper, Van de Bossche, & Dochy, 2015), and engagement in TLBs might be expected to increase more than in long-established teams. Additionally, as shown by the control variables, task interdependence and team structure influence TLBs. Multiple variables may affect TLBs differently, depending also on a team's stage of development (Decuyper et al., 2010). The increase in boundary spanning may be accounted for by increased contact with other teams and individuals outside the team when implementing an idea.

The results provide support for the theoretical view that IWB is a chaotic and multidimensional process and must therefore be measured on multiple dimensions. Although the dimensions of IWB were stable in the first half-year, they were unstable in the second. In relation to the means, it is noticeable that opportunity exploration decreased in the second half of the year, perhaps because the project was ending, and the teams were focused on implementation. The difference in predictions between the dimensions of IWB further indicates a chaotic process, where we found both negative and positive relationships that changed over time. As innovation development depends on team members' past, present and future work activities (Messmann & Mulder, 2012), it is a dynamic process in which progress is not predictable and may vary over time. However, idea promotion seems to be a crucial dimension that positively influences engagement in the others, regardless of time. This suggests that idea promotion may provide a useful starting point for understanding and investigating how IWB can be optimally supported.

TLBs as predictors of IWB

The predictions for TLBs and IWB also change over time in line with theoretical considerations and previous cross-sectional results (e.g., Somech & Khalaili, 2014; Walter & Van der Vegt, 2013). These outcomes contribute to the existing research by adding a temporal perspective. Knowledge is clearly a key resource for team-based innovation development, both for developing a team's creative potential and implementing ideas (Zhang et al., 2011). TLBs of relevance to team knowledge generation are therefore essential for IWB. The results indicate that high engagement in knowledge sharing and team reflexivity is essential for IWB

at the beginning of the school year. Knowledge sharing predicted opportunity exploration and idea promotion in the first half-year, indicating that the broader the knowledge base generated by knowledge sharing, the more knowledge can be used for innovation development, and it is easier to develop high engagement in future opportunity exploration and idea promotion. Knowledge sharing also predicted idea realisation at the end of the year, confirming that teams that share knowledge throughout the year are better in evaluating alternative solutions during idea realisation (Somech & Drach-Zahavy, 2007).

Team reflexivity in early team work seems essential for all dimensions of IWB at the end of the task. That includes making implicit knowledge available to all team members (Müller et al., 2009), processing knowledge and extracting knowledge that is essential for the given task. Continuous reflection on strategies, methods and goals also leads to more advanced action plans for use during subsequent idea promotion and idea realisation (Kolodner, 1997), as unexpected challenges can be avoided, and the best strategy can be selected for idea realisation. Team reflexivity at the outset leads to constructive communication that generates input for future idea generation and opportunity exploration (Buljac-Samardzic & Van Woerkom, 2015).

Reflective teams are also able to recognise knowledge deficits and can cross the team's boundaries to gather information from outside the team. This aligns with our results in relation to boundary spanning, which seems important in the second half of the year. If knowledge of relevance to innovation development is missing, the team can gather the required information from external sources. However, the knowledge available in the team must first be shared and explicated (Müller et al., 2009) for reflection to identify crucial missing information. Boundary spanning also enriches the perspectives of team members, enabling the team to identify opportunities for innovation or how ideas can be adapted, if necessary, for successful realisation.

Taken together, results of this study provide insight into how TLBs and IWB can change over time and which TLBs can foster which dimension of IWB at different time points. The results indicate that at the beginning of teamwork other TLBs are important to reach high engagement in IWB than in later phases of teamwork. It is important to note that also other TLBs than the three examined ones might influence IWB at specific time points. The dynamics in the relationships between TLBs and IWB between T1 and T2 as well as T2 and T3 emphasises the importance of longitudinal studies with more than two waves.

Influence of TLBs and IWB

Engagement in TLBs is influenced by many variables, and the results regarding the control variables indicate that strong task interdependence fosters TLBs; the more team members depend on each other, the more interaction is needed to complete tasks (Rupperecht, Strasser, Mulder, & Gruber, 2011). In addition, a clear team structure seems important, where tasks are distributed among team members and there is a hierarchy within the team. Well-structured teams are able to understand who is responsible for what and who knows what when knowledge is necessary for accomplishing the task (Bunderson & Boumgarden, 2010).

The positive cross-sectional correlations indicate more common influences on the variables and cannot be explained by the autoregressive or cross-lagged paths of the model (Geiser, 2012). Influences may include situational circumstances such as

emotions (Walter & Van der Vegt, 2013). The residual variances of the variables also indicate further influences. In addition to situational circumstances, many variables at different levels may influence TLBs and IWB, as shown by previous studies (cf. Decuyper et al., 2010). For example, there is evidence of positive relationships between motivational constructs and both TLBs and IWB (e.g., Hirst, Van Knippenberg, & Zhou, 2009).

Limitations and implications for future research

A first limitation is that although our results represent a crucial step for further research, they are just a first glimpse of the dynamic relationship between TLBs and IWB. This study was necessary to get a first insight in the reciprocal relationships between TLBs and IWB and to understand how these relationships can change over time. As shown by this study, both TLB and IWB are complex phenomena with no clear pattern in their interconnections, which in addition are affected by many variables at different levels (Decuyper et al., 2010; Messmann & Mulder, 2012). To understand the relationship between TLB and IWB over time more detailed and to capture the complexity of the dynamic relationship a deep and rich description is necessary (Kozlowski, 2015). Thus, qualitative or mixed-method studies can help to get an in-depth understanding of the relationship's time scales and variance and to grasp the complexity of the dynamic relationship between different influences (Raes et al., 2015). Thereby this study provides a starting point for qualitative in-depth studies. Longitudinal studies with more than three waves would make it easier to identify nonlinear dynamics and turning points in behavioural change.

A second limitation could relate to the domain specificity of results. Because of the selection criteria for this sample, (i.e., interdisciplinary teams working on complex and knowledge-intensive tasks to meet challenges), the outcomes reported here have application to similar teams – for instance, industrial research and development teams (Hirst & Mann, 2004). The selected teams' special characteristics, however, could differ from those of teams in other domains. While the requirement to complete tasks in one year may seem adequate for innovation development, and team characteristics remained relatively stable over that period, longitudinal studies are needed to examine contextual change and team development in different domains.

Finally, while the present study focuses exclusively on cognitive aspects of the relationship between TLB and IWB, previous research indicates many different influences, involving both cognitive and affective aspects (e.g., Bunderson & Boumgarden, 2010; Walter & Van der Vegt, 2013). Empirical research linking both elements will be needed to better understand how other variables, such as affects, influence the relationship between TLB and IWB and their mutual influence.

Practical implications

Because innovation is crucial in meeting the demands of societal change, competition and customer expectations, employees' IWB is a key resource for organisations (Anderson et al., 2014). The present findings indicate that idea promotion offers a starting point for fostering opportunity exploration, idea generation and idea realisation. To enrich the team, its members

should communicate with colleagues from outside the team and with other organisations from different domains introducing diverse perspectives that will help to create new knowledge and generate new ideas (cf. Dolinska, 2015). In this way, the team can also win support for implementing their idea.

As our findings identified TLBs as predictors of future IWB, it seems important to raise awareness among organisations and their employees of the ongoing importance of TLBs for developing innovations rather than only at specific times. A high level of engagement in team reflexivity and knowledge sharing at the beginning of a task seems essential for high engagement in IWB. Team members must therefore share their knowledge at the beginning of the task to ensure that the team can exploit this broad knowledge base throughout the process of innovation development. Team knowledge sharing can be supported by creating a climate that team members feel safe for interpersonal risk-taking (Staples & Webster, 2008). Our results also indicate that a team should be organised in a way that ensures that its members need each other to accomplish their tasks. More interdependence means more interaction, fostering open communication for knowledge sharing.

Teams that reflect frequently on their knowledge, strategies and goals build a solid foundation that avoids problems in progressing towards completion. Ensuring that the team is heterogeneous, with diverse expertise and responsibilities, can enrich team reflexivity, and both the team's leader and its members should encourage discussion of critical opinions, especially at the beginning of the task.

Later during the year, boundary spanning also fosters IWB, providing necessary information that is missing within the team. Inviting experts or collaborating with other teams within the organisation can introduce valuable new information. Moreover, it can provide multiple perspectives and diverse opinions for further discussion, in turn fostering further team reflexivity.

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